3. Loops

1. Write a program to print “Bright IT Career” ten times using for loop

Answer:

```
public class PrintText {
    public static void main(String[] args) {
        for (int i = 1; i <= 10; i++) {
            System.out.println("Bright IT Career");
        }
    }
}
```

2. Write a java program to print 1 to 20 numbers using the while loop.

Answer:

```
public class WhileNumbers {
    public static void main(String[] args) {
        int i = 1;

        while (i <= 20) {
            System.out.println(i);
            i++;
        }
    }
}
```

3. Program to equal operator and not equal operators

Answer:

```
public class EqualNotEqual {
    public static void main(String[] args) {
        int a = 10;
        int b = 20;

        System.out.println("a == b : " + (a == b));
        System.out.println("a != b : " + (a != b));
    }
}
```

4. Write a program to print the **odd** and **even** numbers.

Answer:

```
public class OddEven {
    public static void main(String[] args) {

        System.out.println("Even Numbers:");
        for (int i = 1; i <= 20; i++) {
            if (i % 2 == 0) {
                System.out.println(i);
            }
        }

        System.out.println("Odd Numbers:");
        for (int i = 1; i <= 20; i++) {
            if (i % 2 != 0) {
```

```
                System.out.println(i);
            }
        }
    }
}
```

5. Write a program to print largest number among three numbers.
Answer:
```
public class LargestThree {
    public static void main(String[] args) {
        int a = 10, b = 50, c = 30;

        if (a > b && a > c) {
            System.out.println("Largest Number: " + a);
        } else if (b > c) {
            System.out.println("Largest Number: " + b);
        } else {
            System.out.println("Largest Number: " + c);
        }
    }
}
```

6. Write a program to print even number between 10 and 100 using while
Answer:
```
public class EvenNumbers {
    public static void main(String[] args) {
        int i = 10;

        while (i <= 100) {
            if (i % 2 == 0) {
                System.out.println(i);
            }
            i++;
        }
    }
}
```

7. Write a program to print 1 to 10 using the do-while loop statement.
Answer:
```
public class DoWhileExample {
    public static void main(String[] args) {
        int i = 1;

        do {
            System.out.println(i);
            i++;
        } while (i <= 10);
    }
}
```

8. Write a program to find Armstrong number or not

Answer:

```
public class Armstrong {
    public static void main(String[] args) {
        int number = 153;
        int original = number;
        int sum = 0;

        while (number != 0) {
            int digit = number % 10;
            sum = sum + (digit * digit * digit);
            number = number / 10;
        }

        if (sum == original) {
            System.out.println("Armstrong Number");
        } else {
            System.out.println("Not an Armstrong Number");
        }
    }
}
```

9. Write a program to find the prime or not.

Answer:

```
public class PrimeNumber {
    public static void main(String[] args) {
        int number = 13;
        boolean isPrime = true;

        if (number <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i < number; i++) {
                if (number % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            System.out.println(number + " is Prime");
        } else {
            System.out.println(number + " is Not Prime");
        }
    }
}
```

10. Write a program to palindrome or not.

Answer:

```
public class Palindrome {
    public static void main(String[] args) {
        int number = 121;
```

```
        int original = number;
        int reverse = 0;

        while (number != 0) {
            int digit = number % 10;
            reverse = reverse * 10 + digit;
            number = number / 10;
        }

        if (original == reverse) {
            System.out.println("Palindrome Number");
        } else {
            System.out.println("Not a Palindrome Number");
        }
    }
}
```

11. Program to check whether a number is EVEN or ODD using switch
Answer:

```
public class EvenOddSwitch {
    public static void main(String[] args) {
        int number = 15;

        switch (number % 2) {
            case 0:
                System.out.println("Even Number");
                break;

            case 1:
                System.out.println("Odd Number");
                break;
        }
    }
}
```

12. Print gender (Male/Female) program according to given M/F using switch
Answer:

```
public class GenderSwitch {
    public static void main(String[] args) {
        char gender = 'M';

        switch (gender) {
            case 'M':
            case 'm':
                System.out.println("Male");
                break;

            case 'F':
            case 'f':
                System.out.println("Female");
                break;

            default:
```

```
                System.out.println("Invalid Input");
        }
    }
}
```

13. Program for multiple if else statement(Largest number in 10,20 and 30)
Answer:

```
public class MultipleIfElse {
    public static void main(String[] args) {
        int a = 10, b = 20, c = 30;

        if (a > b && a > c) {
            System.out.println(a + " is Largest");
        } else if (b > a && b > c) {
            System.out.println(b + " is Largest");
        } else {
            System.out.println(c + " is Largest");
        }
    }
}
```